

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO2 ASSESSMENT TOOL 1 – Scenario-Based Requirement Analysis

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO2	Assessment:	Assessment Tool 2 – Scenario-Based Requirement Analysis
Weightage:	20%	Total Marks:	25
CO2 Statement:	<i>Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills</i>		
Student Name:		Reg. No.:	
Date:		Duration:	60 minutes

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Annexure A – Scenario-Based Requirement Analysis

Instructions to Students:

Each student will be assigned an individual software project problem statement. Analyze the given scenario and prepare a complete Software Requirement Analysis document by following the steps below.

Question

Based on the **assigned problem statement**, perform a **Scenario-Based Requirement Analysis** and prepare a Software Requirement Specification (SRS) document. Your analysis should include the following:

1. Study and Analyze the Scenario

- Carefully understand the given problem statement.

The Smart Warehouse Management System (SWMS) belongs to the Supply Chain and Logistics Management domain. It is designed to automate warehouse activities such as inventory management, order processing, stock tracking, barcode/RFID scanning, shipment management, and report generation. The system aims to improve operational efficiency, reduce manual work, minimize errors, and provide real-time monitoring of warehouse operations.

- Identify the business domain, stakeholders, existing challenges, and system objectives.

Business Domain:

- Supply Chain Management
- Logistics Management
- Warehouse Management
- Inventory Control
- Distribution and Retail Operations

Primary Stakeholders

- Warehouse Manager
- Warehouse Staff
- Inventory Manager
- Customers
- Suppliers
- Delivery Personnel
- System Administrator

Secondary Stakeholders

- Company Management
- Barcode Scanner System
- RFID Devices
- Database Administrator
- Email Notification Service

Identify Existing Challenges

- Manual inventory management
- Stock mismatch and inaccurate records
- Delayed order processing
- Human errors during stock updates
- Difficulty locating products
- Overstocking and understocking
- Lack of real-time inventory tracking
- Poor warehouse utilization
- Slow report generation
- Limited warehouse visibility

1(e). Identify System Objectives

- Automate warehouse operations
- Improve inventory accuracy
- Enable real-time stock monitoring
- Reduce manual errors

2. Identify Functional and Non-Functional Requirements

- List all functional requirements required by the proposed system.

Functional Requirements

User Management

- User Registration
- User Login
- Logout
- Change Password
- Role-Based Access Control \

Inventory Management

- Add Product
- Edit Product
- Delete Product
- View Product Details
- Categorize Products
- Search Products
- Update Stock Quantity

Warehouse Operations

- Stock In
- Stock Out
- Stock Transfer
- Stock Adjustment
- Inventory Verification

Barcode/RFID Module

- Scan Barcode
- Read RFID Tags
- Update Inventory Automatically

Order Management

- Create Order
- Modify Order
- Cancel Order
- Track Order Status

Supplier Management

- Add Supplier
- Edit Supplier
- Delete Supplier
- View Supplier Details

Shipment Management

- Schedule Shipment
- Dispatch Orders
- Track Delivery
- Confirm Delivery

Reporting Module

- Daily Inventory Report
- Monthly Report
- Sales Report
- Low Stock Report

Notification Module

- Low Stock Alert
- Order Status Notification
- Delivery Notification
- Expiry Alert
 - Identify suitable non-functional requirements such as performance, security, reliability, usability, scalability, maintainability, and availability.

Performance

- System response time below 3 seconds.
- Handle at least 500 concurrent users.

Security

- Secure user authentication
- Password encryption
- Role-based authorization
- Data encryption
- Secure backups

Reliability

- 99.9% uptime
- Automatic recovery from failures
- Regular backups

Usability

- User-friendly interface
- Easy navigation
- Responsive design
- Mobile compatibility

Scalability

- Support multiple warehouses
- Easy database expansion
- Cloud deployment support

Maintainability

- Modular software architecture
- Easy software updates
- Proper documentation

Availability

- 24×7 availability
- Cloud hosting support

3. Identify Actors and Use Cases

- Identify all primary and secondary actors interacting with the system.

Primary Actors

- Warehouse Manager
- Warehouse Staff
- Inventory Manager
- Supplier
- Customer
- Delivery Personnel
- Administrator

Secondary Actors

- Barcode Scanner
- RFID Reader
- Payment Gateway
- Email Notification Service
- Database Server

- List the major use cases associated with each actor.

Actor	Major Use Cases
Warehouse Manager	View Dashboard, Generate Reports, Monitor Inventory
Warehouse Staff	Add Stock, Remove Stock, Scan Barcode
Inventory Manager	Verify Inventory, Update Stock
Supplier	Supply Products, Receive Purchase Orders
Customer	Place Orders, Track Orders
Delivery Personnel	Update Delivery Status
Administrator	Manage Users, Backup Database

- Draw a Use Case Diagram representing the interactions between actors and the system.

4.Prepare the Software Requirement Specification (SRS)

Develop an SRS document containing:

- Introduction

The Smart Warehouse Management System (SWMS) is developed to automate warehouse operations, improve inventory management, reduce manual work, and provide real-time monitoring of stock movement. The system enables efficient warehouse management through barcode/RFID technology and automated reporting.

- Problem Description

Traditional warehouse systems rely on manual record keeping, resulting in inventory errors, delayed shipments, poor stock visibility, and inefficient warehouse operations. An automated Smart Warehouse Management System is required to overcome these challenges.

- Scope of the System

The system includes:

- User Management
- Inventory Management
- Warehouse Operations
- Barcode/RFID Scanning
- Order Processing
- Shipment Management

- Functional Requirements

- User Registration and Login
- Product Management
- Inventory Tracking
- Barcode/RFID Scanning
- Order Management
- Supplier Management
- Shipment Tracking
- Report Generation
- Notification System
- Dashboard Management

- Non-Functional Requirements

- High Performance
- High Security
- Reliability
- Availability
- Scalability
- Maintainability
- Usability
- Portability

- Use Case Descriptions

Use Case 1: Login

Actor: User

Precondition

- User must be registered.

Main Flow

1. Enter username.
2. Enter password.
3. Click Login.
4. System validates credentials.
5. Dashboard opens.

Postcondition

- User successfully logs into the system.

Use Case 2: Add Inventory

Actor: Warehouse Staff

Precondition

- Staff logged into the system.

Main Flow

1. Scan barcode.
2. Enter quantity.
3. Save details.
4. Inventory updates automatically.

Postcondition

- Product stock updated successfully.

Use Case 3: Generate Report

Actor: Warehouse Manager

Main Flow

1. Select report type.
2. Select date range.
3. Click Generate.
4. System generates report.

Postcondition

- Report displayed and available for download.
 - Data Requirements

User Table

- User ID
- Username
- Password
- Role

Product Table

- Product ID
- Product Name
- Category
- Quantity

- Barcode
- Price

Supplier Table

- Supplier ID
- Supplier Name
- Contact Number
- Address

Customer Table

- Customer ID
- Customer Name
- Email
- Phone Number

Order Table

- Order ID
- Customer ID
- Product ID
- Quantity
- Order Date
- Status

Shipment Table

- Shipment ID
- Order ID
- Delivery Status
- Dispatch Date

- External Interface Requirements

User Interface

- Login Screen
- Dashboard
- Inventory Page
- Reports Page
- Orders Page

Hardware Interface

- Barcode Scanner
- RFID Reader
- Printer

Software Interface

- MySQL Database
- Email Server
- Payment Gateway

Communication Interface

- Internet
- Wi-Fi
- LAN

- System Features

- Secure Login
- Dashboard
- Inventory Tracking
- Barcode Scanning
- RFID Support
- Low Stock Alerts
- Shipment Tracking
- Supplier Management
- Report Generation
- Cloud Backup

4. Identify Assumptions and Constraints

- List the assumptions considered while developing the system.

Assumptions

- Users are trained to operate the system.
- Stable internet connection is available.
- Barcode/RFID devices function correctly.
- Database server remains operational.
- Products are labeled correctly.

- Identify technical, operational, economic, legal, and time-related constraints affecting the project.

Constraints

Technical Constraints

- MySQL database
- Web-based architecture
- Barcode/RFID hardware compatibility

Operational Constraints

- Warehouse staff must update stock promptly.
- Regular maintenance is required.

Economic Constraints

- Limited project budget
- Hardware procurement costs
- Software licensing expenses

Legal Constraints

- Compliance with data protection regulations
- Secure storage of customer and supplier data

Time Constraints

- Development must be completed within the allocated schedule.
- Testing and deployment should finish before production use.

5. Validate Requirement Completeness and Consistency

- Verify whether all stakeholder requirements have been addressed.

Verify Stakeholder Requirements

- Warehouse Manager requirements addressed.
 - Warehouse Staff requirements addressed.
 - Inventory Manager requirements addressed.
 - Supplier requirements addressed.
 - Customer requirements addressed.
 - Delivery Personnel requirements addressed.
 - Administrator requirements addressed.
- Check for ambiguity, redundancy, inconsistency, and missing requirements.

Ambiguity Check

- All requirements are clearly defined.
- User roles and responsibilities are specified.

Redundancy Check

- Duplicate requirements have been removed.
- Each feature is defined only once.

Consistency Check

- Functional requirements align with system objectives.
- Use cases match the identified actors.
- Data requirements support all modules.

Missing Requirements Check

- No major functional or non-functional requirements are missing.
- All system modules have been covered.
 - Suggest improvements to enhance the quality and completeness of the requirement specification.
- Integrate AI-based inventory demand forecasting.
- Add IoT sensors for real-time warehouse monitoring.
- Develop a mobile application for warehouse staff.
- Enable predictive analytics for stock replenishment.
- Support QR code scanning in addition to barcode/RFID.
- Add multilingual user interface support.
- Implement cloud-based backup and disaster recovery.
- Provide customizable dashboards and advanced analytics.

Rubrics for Calculation

Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Scenario Understanding and Problem Analysis (30%)	Demonstrates thorough understanding of the scenario, stakeholders, objectives, and business context with clear analysis.	Demonstrates good understanding with minor omissions.	Basic understanding with limited analysis.	Poor understanding; major aspects missing or incorrect.
Functional and Non-Functional Requirement Identification (25%)	Clearly identifies comprehensive, accurate, and well-classified functional and non-functional requirements.	Identifies most requirements with minor classification errors.	Identifies only basic requirements; several omissions.	Requirements are incomplete, inaccurate, or poorly classified.
Actor Identification	Correctly identifies all relevant actors and	Most actors and use cases	Limited actors/use cases	Actors/use cases missing or

and Use Case Modelling (20%)	use cases; use case diagram is complete, accurate, and follows UML standards.	identified; diagram has minor errors.	identified; diagram partially correct.	incorrect; diagram absent or inaccurate.
Software Requirement Specification (SRS) Documentation (15%)	SRS is well-structured, complete, professionally organized, and includes all required sections.	SRS is organized with minor missing sections or formatting issues.	SRS includes only essential sections with limited detail.	SRS is incomplete, poorly organized, or lacks essential components.
Assumptions, Constraints, and Requirement Validation (10%)	Clearly identifies realistic assumptions and constraints; validates requirements for completeness, consistency, and ambiguity with appropriate justification.	Identifies most assumptions and constraints; validation is adequate.	Limited assumptions/constraints; validation is superficial.	Assumptions, constraints, and validation are missing or incorrect.

Criteria	Mark
Scenario Understanding and Problem Analysis (30%)	/5
Functional and Non-Functional Requirement Identification (25%)	/5
Actor Identification and Use Case Modelling (20%)	/5
Software Requirement Specification (SRS) Documentation (15%)	/5
Assumptions, Constraints, and Requirement Validation (10%)	/5
TOTAL	/25